12/March/26 * EXPERIMENT- 07 *

* AIM :- Determining the probability of a given case using binomial distribution.

* EXPERIMENT :- The mean and variance of binomial distribution are 4 and 4/3 respectively. Find :-

a) Probability of 2 successes

b) Probability of more than 2 successes.

* THEORY :- A random variable X has the binomial distribution if its probability distribution is given by ;

$$\Rightarrow \boxed{P[X = x] = {}^nC_x p^x q^{n-x}}$$

$$, x = 0, 1, 2, \ldots, n$$

$$\Rightarrow \boxed{\text{Mean} = \mu_1' = np}$$

$$\Rightarrow \boxed{\text{Variance} = npq}$$

CALCULATIONS

→ Given ;

$$np = 4 \quad \text{——} \quad (1)$$

$$npq = 4/3 \quad \text{——} \quad (2)$$

Using (1),

$$4q = 4/3 \Rightarrow \underline{q = 1/3}$$

$$p \Rightarrow 1 - q \Rightarrow \underline{2/3}$$

$$n\left(\frac{2}{3}\right) = 4$$

$$\Rightarrow \underline{n = 6}$$

i) $P[X = 2] = {}^6C_2 \left(\frac{2}{3}\right)^2 \left(\frac{1}{3}\right)^4$

$$= \frac{6!}{2!\,4!} \times \frac{4}{9} \times \frac{1}{81}$$

$$= 15 \times \frac{4}{729}$$

$$= \boxed{\frac{20}{243}}$$

ii) $P[X > 2] = 1 - P[X \leq 2]$

$$= 1 - \{ P[X=0] + P[X=1] + P[X=2] \}$$

$$= 1 - \left\{ \frac{1}{729} + \frac{4}{243} + \frac{20}{243} \right\}$$

$$= 1 - \frac{73}{729}$$

$$= \boxed{\frac{656}{729}}$$

* <u>RESULT</u> :-

→ Probability of getting exactly 2 successes is $\frac{20}{243}$.

→ Probability of getting more than 2 successes is $\frac{656}{729}$.

—— x —— x —— x ——